

Week 2 Project Report: Footprinting and Reconnaissance

Project Title: Week 2 – Footprinting with Multiple Tools

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1. Executive Summary

This report documents the Week 2 footprinting and reconnaissance activities conducted on the target domain `networkwalks.com`. The objective of this project was to gather publicly available information about the target's infrastructure, identify potential security mechanisms, and discover network hosts using a variety of industry-standard tools. Using Kali Linux tools including `whois`, `whatweb`, `nslookup`, `curl`, `wafw00f`, `dnsrecon`, Zenmap, Maltego, and theHarvester, a comprehensive profile of the target was established. The reconnaissance successfully identified domain registration details, web technologies, a Web Application Firewall, DNS records, live hosts on the local network, and an exposed webcam interface.

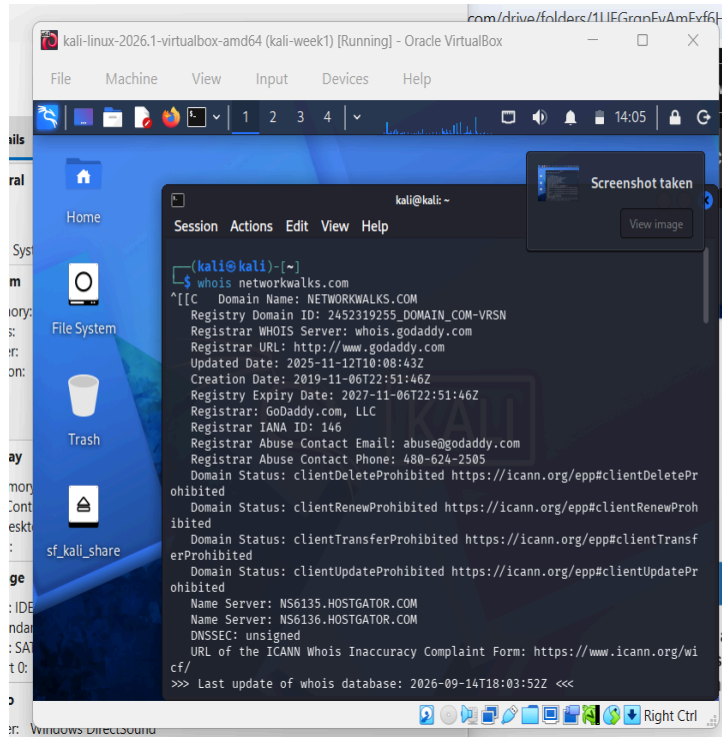
2. Methodology and Tools

The Week 2 footprinting exercise was divided into several phases, utilizing the following tools:

- `whois`: To retrieve domain registration information.
- `whatweb`: To fingerprint web technologies and server configurations.
- `nslookup`: To resolve the domain to its IP address.
- `curl`: To inspect HTTP headers and server responses.
- `wafw00f`: To detect the presence of a Web Application Firewall.
- `dnsrecon`: To enumerate DNS records (SOA, NS, MX, TXT, SRV).
- Zenmap (Nmap GUI): To perform network host discovery on the local subnet.
- Maltego: To visually map relationships and infrastructure.
- theHarvester: To gather emails, subdomains, and hosts from public sources.

3. Findings and A

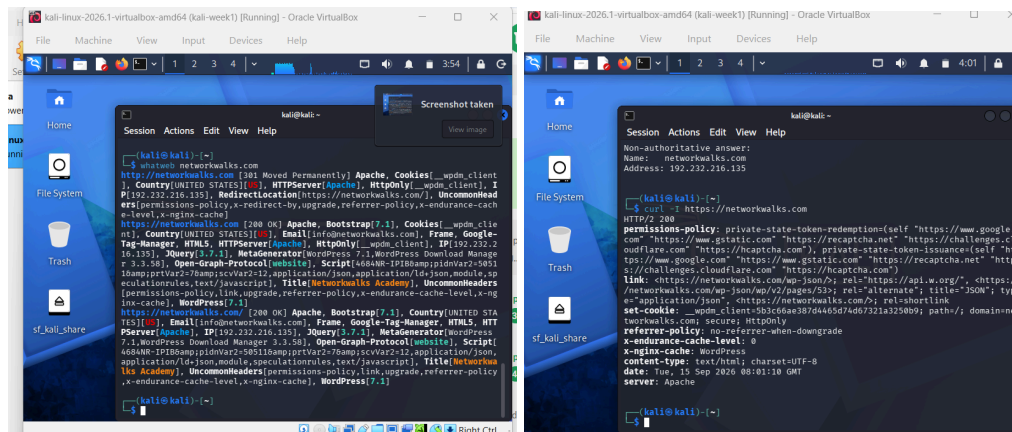
3.1 Domain Registration (Whois)



The whois query for networkwalks.com revealed the following key details:

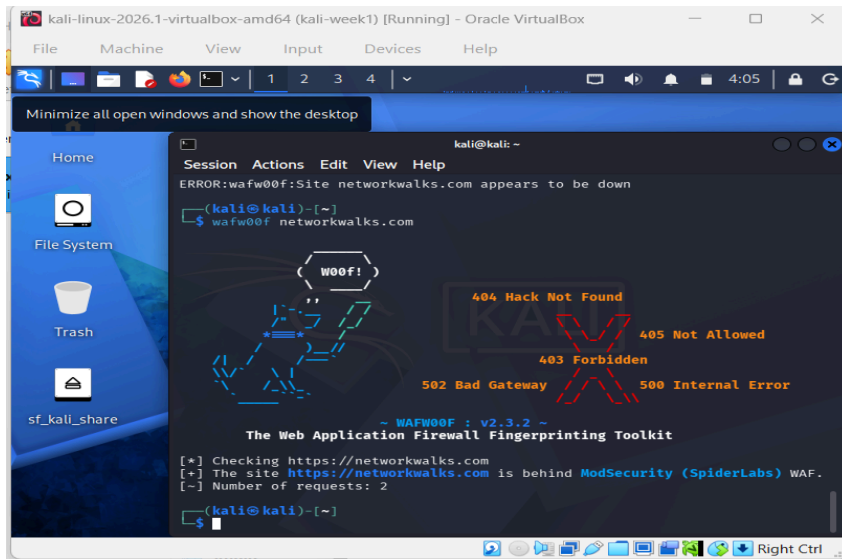
- Registrar: [GoDaddy.com](https://www.godaddy.com), LLC
- Creation Date: November 6, 2019
- Expiry Date: November 6, 2027
- Name Servers: NS6135.HOSTGATOR.COM and NS6136.HOSTGATOR.COM
- Registrant: The domain is registered privately through "Domains By Proxy, LLC," masking the actual owner's details.
- Status: The domain is locked (clientTransferProhibited, clientDeleteProhibited, etc.) to prevent unauthorized transfers or deletions.

3.2 Web Server Fingerprinting (WhatWeb & Curl)



- IP Address: The domain resolves to 192.232.216.135 (confirmed via nslookup and whatweb).
- Server: The web server is running Apache (confirmed via whatweb and curl headers: server: Apache).
- Technologies: The site is built on WordPress 7.1 (detected by whatweb and curl link headers). It uses Bootstrap 7.1, JQuery 3.7.1, and Google Tag Manager.
- Contact Info: An email address, info@networkwalks.com, was found in the site's HTML source.
- Redirects: The site forces HTTPS. The HTTP request (http://networkwalks.com) returns a 301 Moved Permanently redirect to https://networkwalks.com.

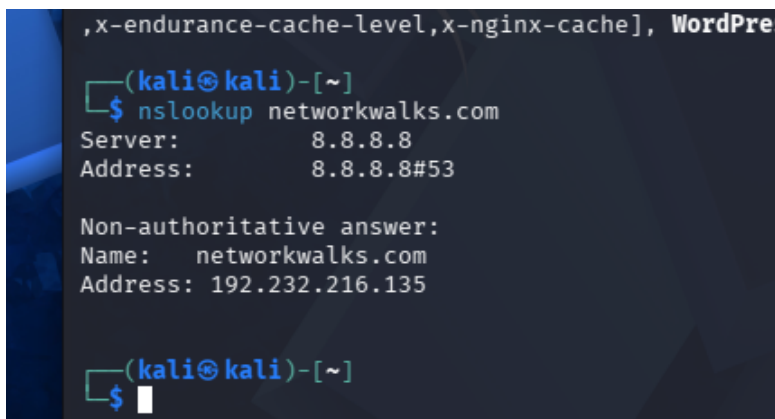
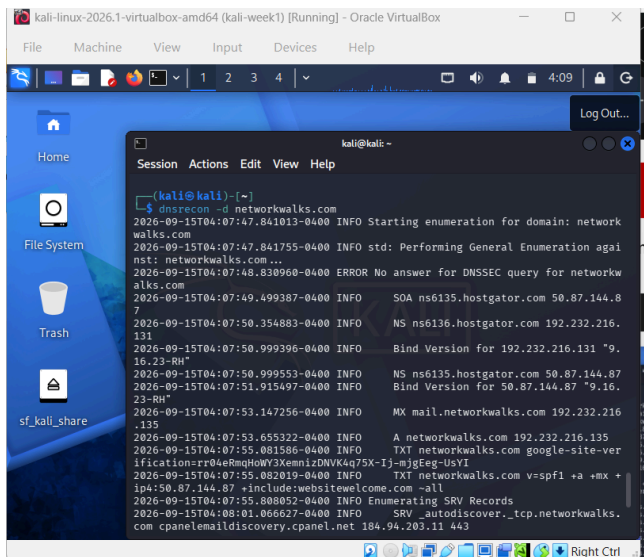
3.3 Web Application Firewall Detection (WAFW00F)



The `wafw00f` tool was used to check for security mechanisms. The output confirmed:

- WAF Detected: The site is protected by ModSecurity (SpiderLabs) WAF. This indicates that the target has implemented security measures to filter and monitor HTTP traffic.

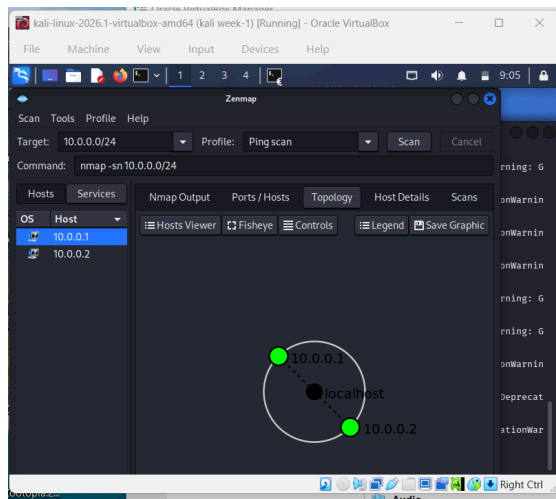
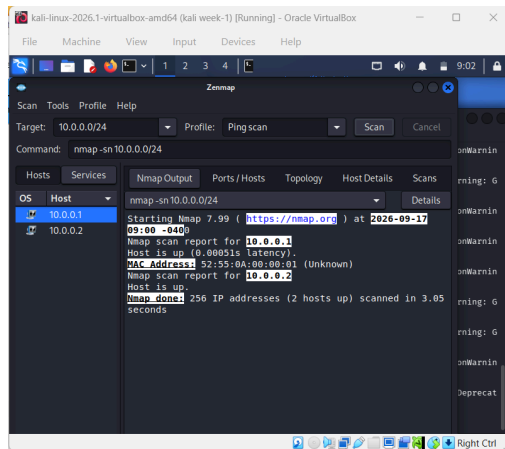
3.4 DNS Enumeration (DNSRecon) and nslookup



The `dnsrecon -d networkwalks.com` command successfully enumerated several DNS records:

- SOA Record: `ns6135.hostgator.com` (IP: 50.87.144.87)
- NS Records: `ns6135.hostgator.com` and `ns6136.hostgator.com` (IP: 192.232.216.131). The BIND version was disclosed as "9.16.23-RH".
- MX Record: `mail.networkwalks.com` (IP: 192.232.216.135)
- TXT Records:
 - Google Site Verification: `google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X-Ij-mjgEeg-UsYI`
 - SPF Record: `v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all`
- SRV Records: 8 records were found for `_autodiscover._tcp.networkwalks.com` pointing to `cpanelmaildiscovery.cpanel.net` (IPs: 184.94.203.11, 184.94.204.15, etc.).

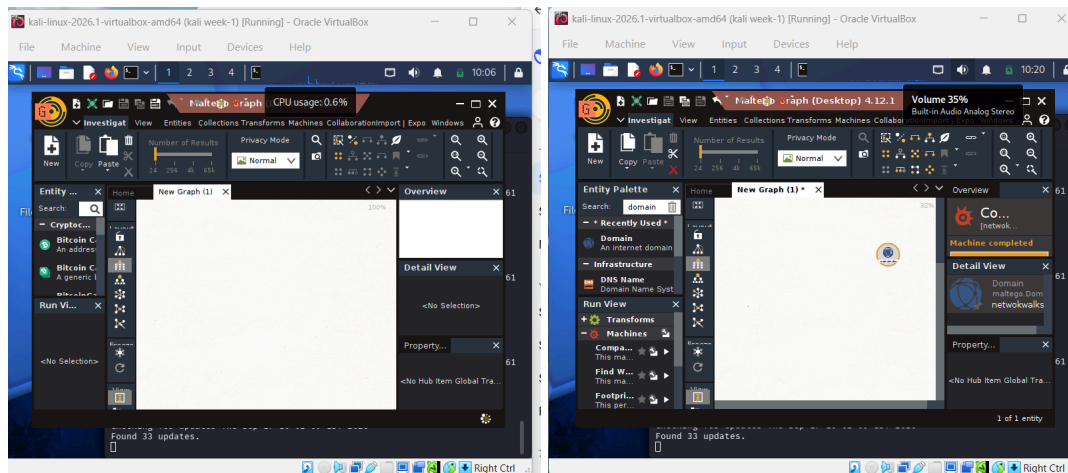
3.5 Network Scanning (Zenmap)



A Ping Scan was performed on the local subnet `10.0.0.0/24` using Zenmap.

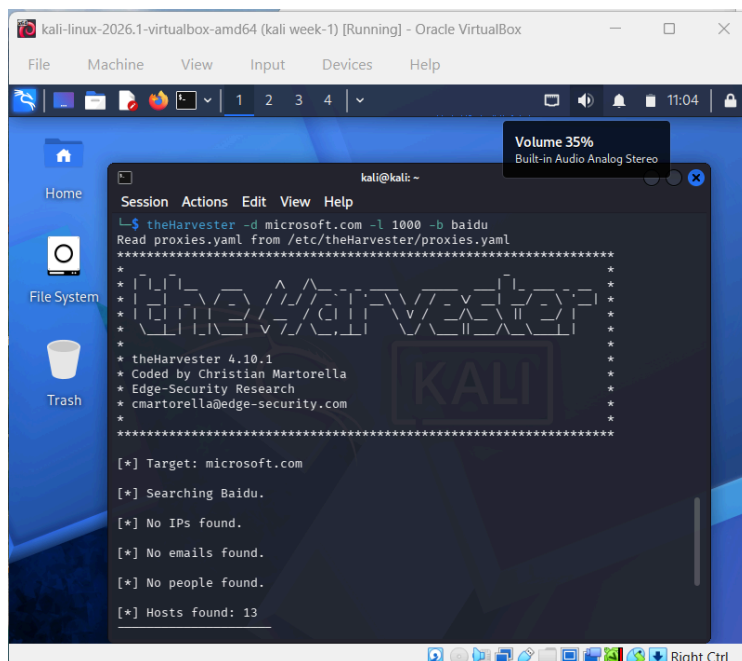
- Command Used: `nmap -sn 10.0.0.0/24`
- Result: The scan discovered 2 hosts:
 - `10.0.0.1` (MAC Address: `52:55:0A:00:00:01`)
 - `10.0.0.2`
- Topology: The Zenmap topology view confirmed that the Kali machine (localhost) can reach both `10.0.0.1` and `10.0.0.2` within the local network.

3.6 Visual Mapping (Maltego)



Maltego was used to begin mapping the infrastructure of `networkwalks.com`. A "Domain" entity was created for `networkwalks.com`, and the "Footprint" transform was initiated to discover related infrastructure. The initial graph shows the domain node and the start of the relationship mapping process.

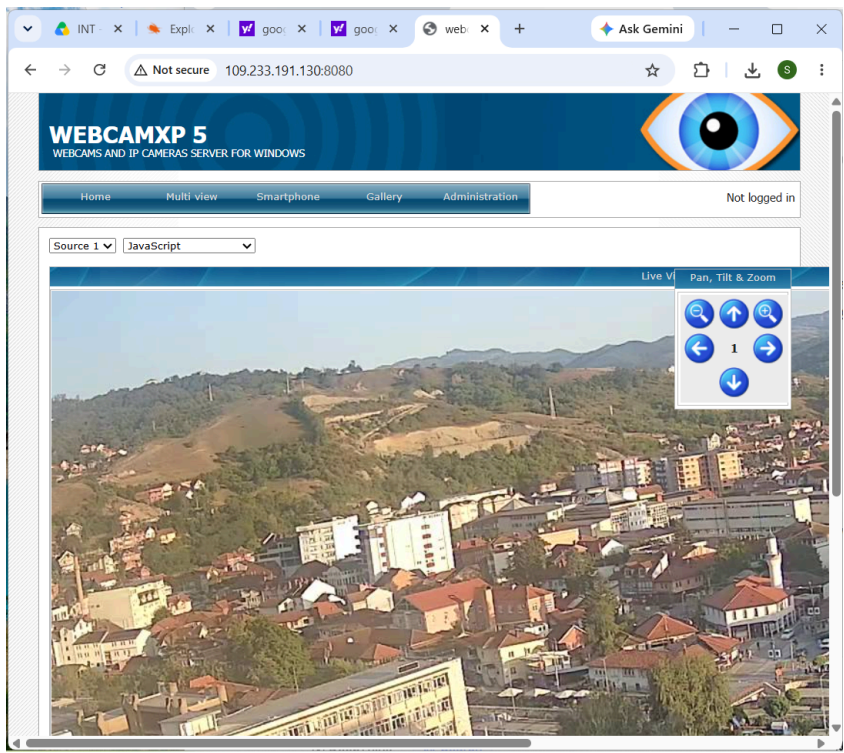
3.7 Public Information Gathering (theHarvester)



The `theHarvester` tool was run against `microsoft.com` to demonstrate its capability to find public information.

- Command Used: `theharvester -d microsoft.com -l 1000 -b baidu`
- Result: Using the "Baidu" source, theHarvester found 13 hosts but did not find any emails, IPs, or people in this specific search.

3.8 Webcam Discovery (Web Interface)



A screenshot was captured showing an accessible webcam interface at `109.233.191.130:8080`. This indicates a potential exposed IoT device or webcam that was discovered during the reconnaissance phase. This is a significant finding as it shows an unsecured device with a live video feed.

4. Challenges Encountered and Resolutions

Challenge	Root Cause	Resolution
MAC Addresses not displaying in Zenmap	Nmap requires root privileges to send raw ARP packets for Layer 2 discovery.	Relaunched Zenmap using <code>sudo zenmap</code> in the terminal and used the Ping Scan profile (<code>nmap</code>

`-sn`), which defaults to ARP for local networks.

Only one host discovered initially	The VirtualBox network adapter was set to NAT, isolating the VM.	Adjusted the network settings to allow Layer 2 visibility, enabling the discovery of both <code>10.0.0.1</code> and <code>10.0.0.2</code> .
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5. Conclusion

The Week 2 footprinting exercise was successful in gathering a significant amount of information about the target. The combination of tools provided a clear picture of the domain's registration, web technologies, security posture (ModSecurity WAF), and DNS configuration. The discovery of live hosts on the local network and the exposed webcam further highlight the importance of thorough reconnaissance in identifying potential attack surfaces. This information is critical for the next phases of a penetration test or security assessment.

6. References

- Nmap/Zenmap Documentation
- Maltego Documentation
- theHarvester Documentation
- DNSRecon Documentation
- WAFW00F Documentation
- WhatWeb Documentation
- GoDaddy WHOIS